

Normalization Process for Refinement of Database

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How to design the database?

There are two approaches

- E-R Modeling:

- ❑ Identifying entity and relations
- ❑ Conversion of ER diagram to tables
- ❑ Top down approach

- Normalization:

- ❑ Refinement of database
- ❑ Design database by using tests or normal forms
- ❑ Bottom up approach



Database Design Techniques

- Top down Approach
 - E R Modeling
- Bottom Up approach
 - Normalization



Operations on Database

- Insertion
- Updation
- deletion
- Retrieval

Normalization

- It is a process of decomposing a larger table into smaller tables so that it satisfy series of tests. If the database satisfies the test, then database is considered normalized according to that test or rule or degree, otherwise database is considered un-normalized.
- When a test fails, the relation violating that test must BS decomposed into relations so that it individually meet the normalization tests.

Process of Normalization

		APPLY TEST		

Process of Normalization

Process of Normalization

	Normalized			

Process of Normalization

Process of Normalization

	Unnormalized			

Process of Normalization

		Unnormalized		

APPLY RULE FOR DECOMPOSITION

Process of Normalization

		Unnormalized		

APPLY RULE FOR DECOMPOSITION

	APPLY TEST	

	APPLY TEST	

Process of Normalization

		Unnormalized		

APPLY RULE FOR DECOMPOSITION

	✓	

	APPLY TEST	

Process of Normalization

		Unnormalized		

APPLY RULE FOR DECOMPOSITION

	Normalized	

	APPLY TEST	

Process of Normalization

Unnormalized

APPLY RULE FOR DECOMPOSITION

Normalized



Process of Normalization

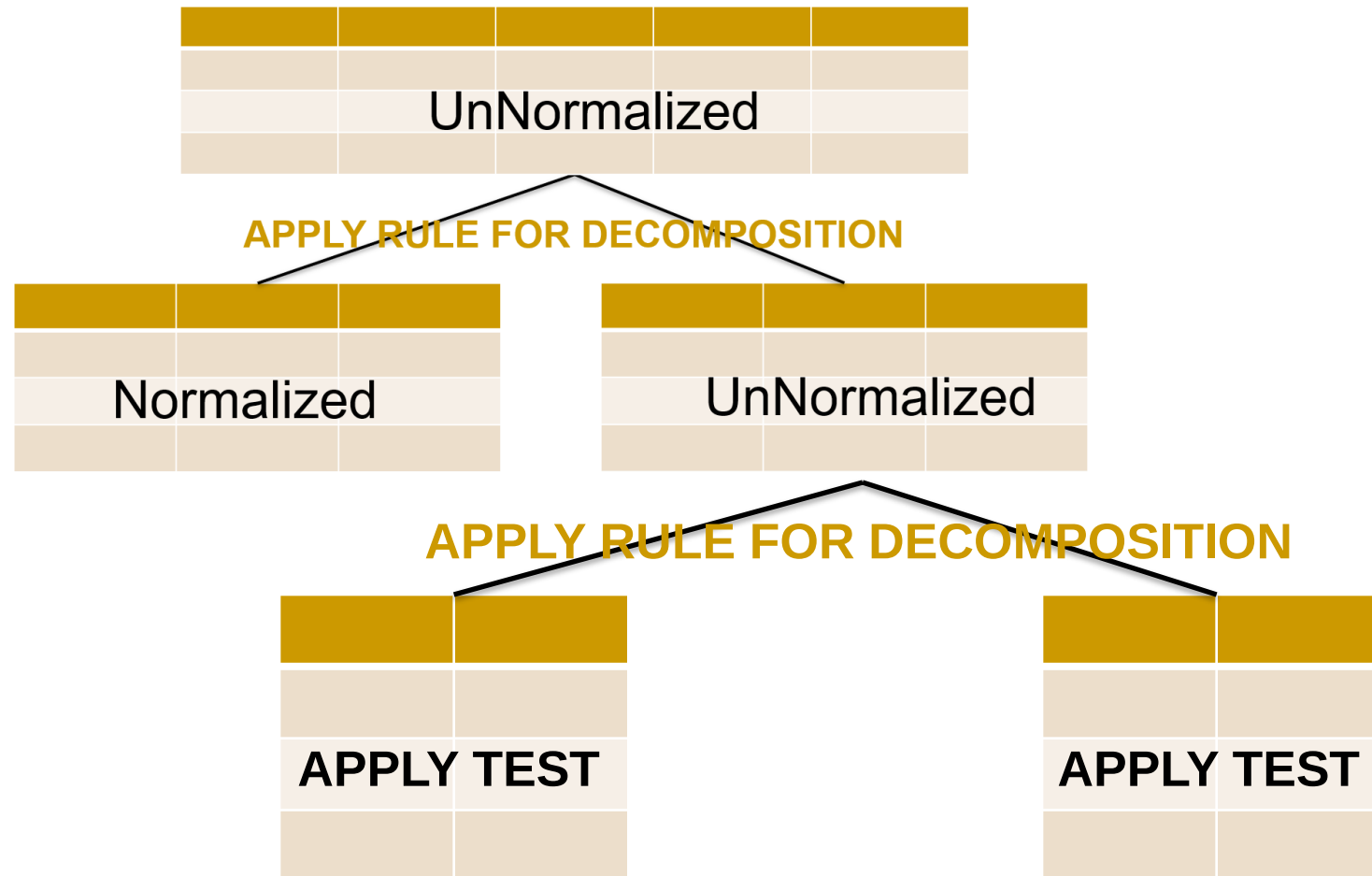
		Unnormalized		

APPLY RULE FOR DECOMPOSITION

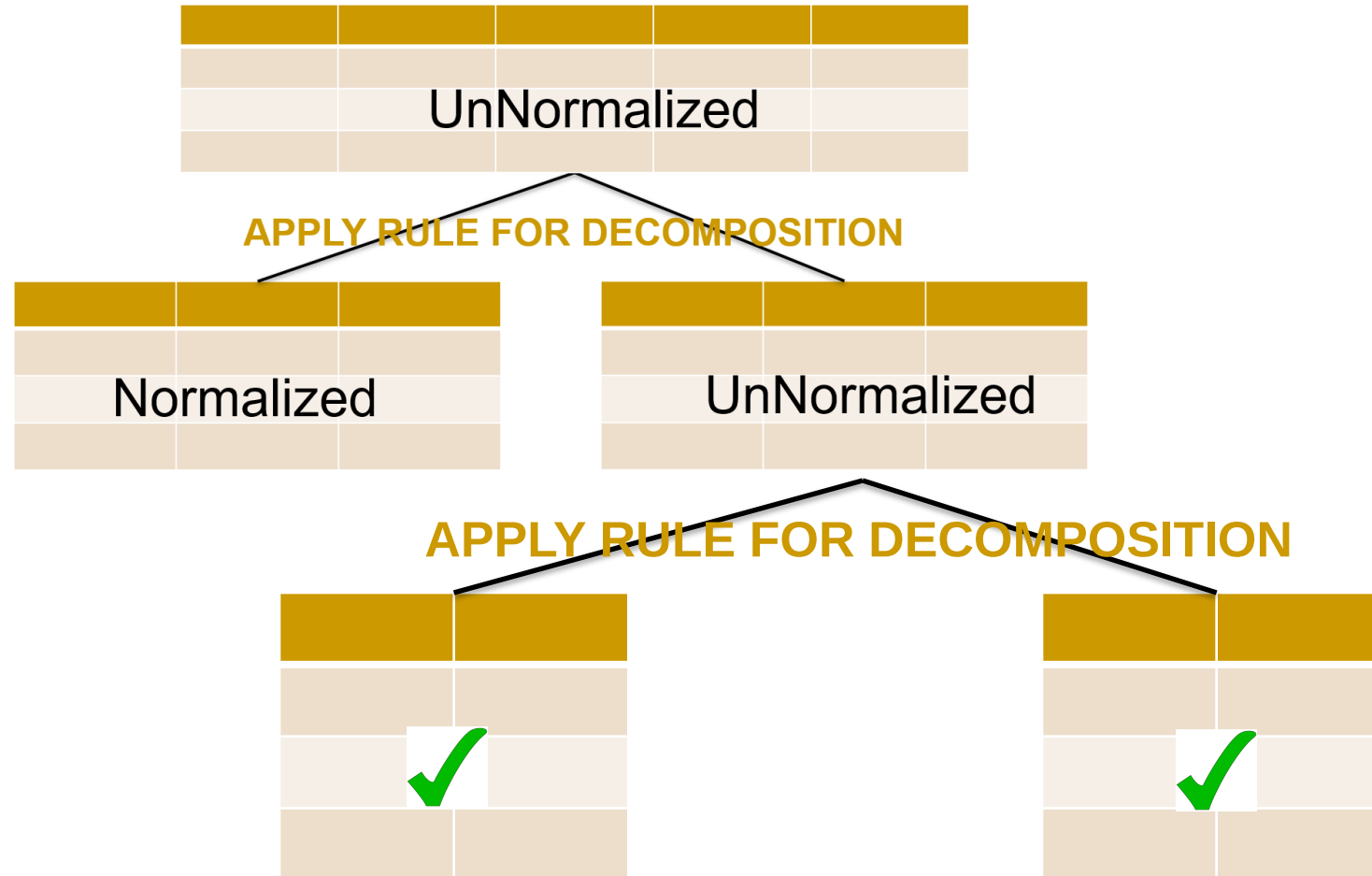
	Normalized	

	Unnormalized	

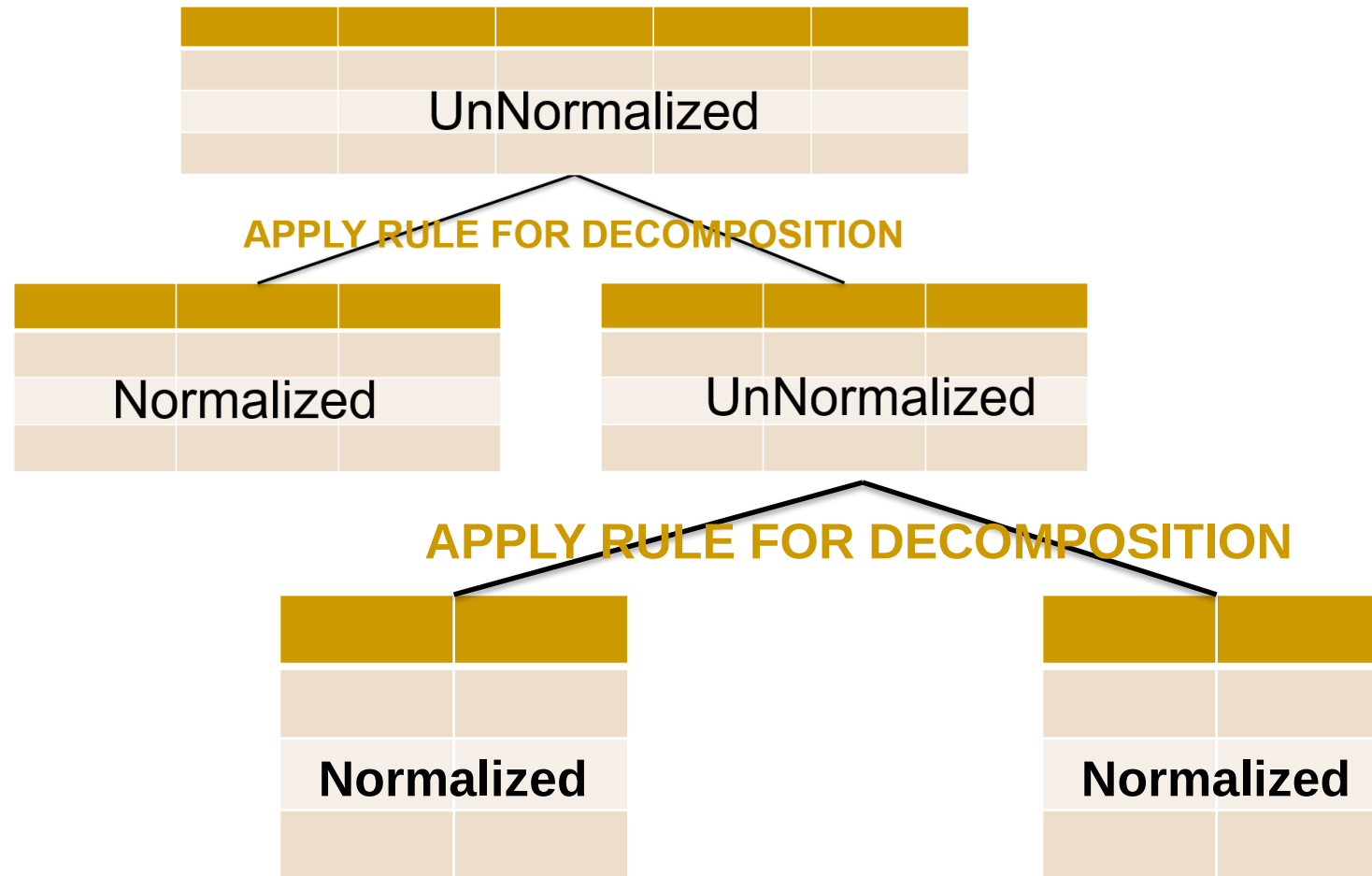
Process of Normalization



Process of Normalization



Process of Normalization



Normalization

- There are five series of test that we apply on the database, so there are five degree or rules of normalization which are known as Normal Forms.
- For example: First Normal Form, Second Normal Form and so on.

Objectives of Normalization

Objectives of Normalization

- To create a formal framework to verify that designed database is accurate and it ensures that we can perform insertion, update, delete and retrieve operation without any anomalies or issues.
- To have a formal framework normalization process involves series of test to verify that our designed database is accurate.
- Here, we analyze the relations based on their keys and different kind of dependencies among their attributes.

Objectives of Normalization

- To reduce the need for restructuring the relations as new data types are introduced.
 - To obtain powerful relational retrieval algorithms.
 - In simple words, it assure that our designed database is correct and we can start building our system over it.
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Objectives of Normalization

- For this we carry out series of tests on individual relation schema so that the relational database can be normalized to some degree.
 - When a test fails, the relation violating that test must be decomposed into relations that individually meet the normalization tests.
-

Normal Forms

First N/F	Second N/F	Third N/F	Fourth N/F	Fifth N/F
Eliminate Repeating Groups	Non-key attribute Fully Functional Dependence on PK	Eliminate Transitive Dependence on PK	Remove Multi Value Dependence	Projection Join Dependency

Normalization

Concept of Functional Dependence

Functional Dependence (FD)

- In a relation R having two attributes X and Y and if for each value of X there is only one value of Y, then Y is called functionally dependent on X.
- This is represented as $X \rightarrow Y$.
- Here, X is the determinant and Y is the determined or *X functionally determines Y*.
- In other words, *Y is functionally determined by X*.
- **For each value of the determinant there should be associated one and only one value of the determined.**

Functional Dependence (FD): Example

ID	Name	Class
1	Sam	BS
2	Nik	BS
3	Watson	MS
4	Tim	BS
5	Sam	BS
7	Tim	MS

ID $\rightarrow$ Name

ID $\rightarrow$ Class

Name $\nrightarrow$ ID

Class $\nrightarrow$ ID

Functional Dependence (FD): Example

ID	Name	Class
<u>1</u>	Sam	BS
<u>2</u>	Nik	BS
3	Watson	MS
<u>4</u>	Tim	BS
<u>5</u>	Sam	BS
7	Tim	MS

Unique

ID $\rightarrow$ Name ✓

ID $\rightarrow$ Class ✓

Name $\rightarrow$ ID

Class $\rightarrow$ ID

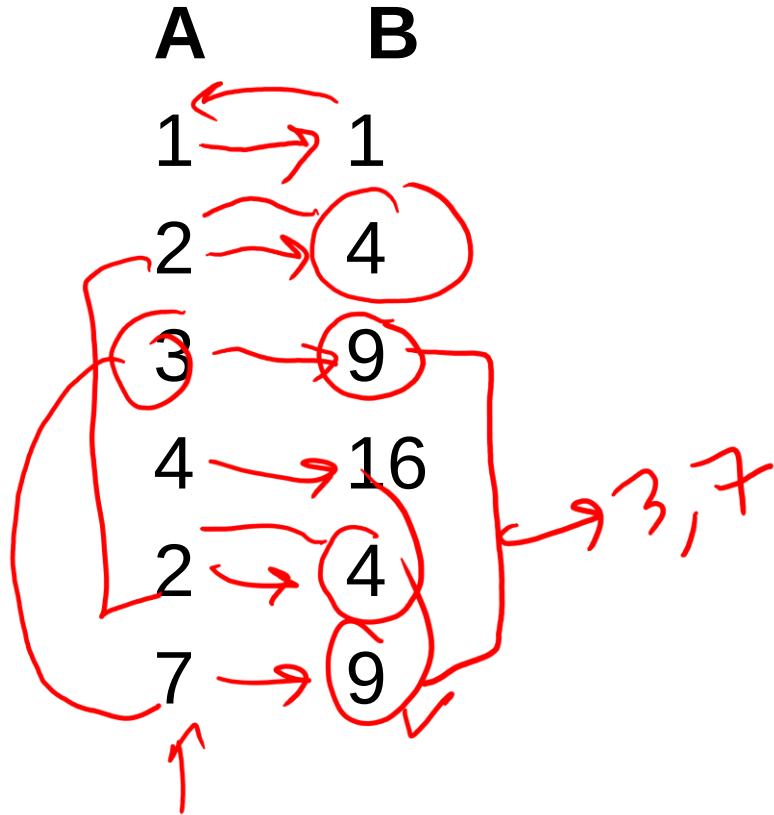
Functional Dependence (FD): Example

A	B
1	1
2	4
3	9
4	16
2	4
7	9

$A \rightarrow B$

$B \nrightarrow A$

Functional Dependence (FD): Example



$A \rightarrow B$ ✓

$\bar{B} \rightarrow A$

$B \nrightarrow A$

Functional Dependence (FD): Example

A	B
1	1
2	4
3	9
4	16
3	10

$A \rightarrow B$

$B \rightarrow A$

Functional Dependence (FD): Example

A	B
1	1
2	4
3	9
4	16
3	10

↑
unique

$A \not\rightarrow B$

$B \rightarrow A$ ✓

Functional Dependence (FD): Example

SNO	SNAME	CITY	STATUS
S1	Sam	Pullman	10
S2	ram	Pullman	10
S3	Sam	Spokane	20
S4	Nik	Seattle	10

$SNO \rightarrow SNAME$ $SNAME \nrightarrow SNO$
 $SNO \rightarrow CITY$ $CITY \nrightarrow SNO$
 $SNO \rightarrow STATUS$ $STATUS \nrightarrow SNO$

Rule: Every primary key of relation functionally determines all other non-key attributes of the relation.

Normalization

Understanding the Dataset for Computer Center Case Study

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
C3	C++	KJP	101	A2	P-II	30	2
			106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
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			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every Course_Code there is one Course_Name
- For every Course_Code there is one Teacher_Name

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
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			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every Course_Code there are many students or RollNo
- For every Course_Code there are many values of Name

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every Course_Code there are many values of System_Used
- For every Course_Code there are many values of Hourly_Rate, Total_Hrs

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
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			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every RollNo there is only one value of Name
- For every RollNo there is only one value of System_Used

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
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			102	A3	Celeron	10	6
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			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every RollNo there is only one value of Hourly_Rate
- For every RollNo there are multiple values of Total_Hrs

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For every Course_Code and RollNo combination, there is one value of Total_Hrs

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
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			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

- For Course_Code, Course_Name and Teacher_Name group there are multiple values of RollNo, Name, System_Used, Hourly_Rate and Total_Hrs

Understanding Dataset

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
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C4	Java	Kumar	109	A10	Cyrix	20	2

■ Non-Repeating Group

Course_Code,
Course_Name and
Teacher_Name

■ Repeating Group

RollNo, Name,
System_Used,
Hourly_Rate and
Total_Hrs

Normalization

First Normal Form

First Normal Form

Definition of First Normal Form

- A relation is said to be in First Normal Form (1NF) if and only if every entry of the relation (the intersection of a tuple and a column) has at most a single value.
- In other words “a relation is in First Normal Form if and only if all underlying domains contain atomic values or single value only.”

First Normal Form

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
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First Normal Form

STUDENT (Unnormalized table)

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C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
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			105	A6	P-II	30	1
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			106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Rule to Convert Dataset to First Normal Form

First Approach: Flattening the table

- The first approach known as “flattening the table” removes repeating groups by filling in the “missing” entries of each “incomplete row” of the table with copies of their corresponding non-repeating attributes.

First Approach: Flattening the table

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
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			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
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C4	Java	Kumar	109	A10	Cyrix	20	2

First Approach: Flattening the table

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First Approach: Flattening the table

STUDENT (Unnormalized table)

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C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
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			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

First Approach: Flattening the table

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
			102	A3	Celeron	10	6
			103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
C3	C++	KJP	101	A2	P-II	30	2
			106	A7	P-IV	40	3
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			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

First Approach: Flattening the table

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Rule to Convert Dataset to First Normal Form

Second Approach: Decomposition of the table

- The second approach for normalizing a table requires that the table be decomposed into two new tables that will replace the original table.

Rule of decomposition

- One Table

Create a relation with Table identifier (attribute which uniquely define the table) of the original table and all the non-repeating attributes.

- Second Table

The other table contains a copy of the table identifier and all the repeating attributes.

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
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			103	A4	P-IV	40	1
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			104	A5	P-III	35	3
			105	A6	P-II	30	1
C3	C++	KJP	101	A2	P-II	30	2
			106	A7	P-IV	40	3
			107	A8	P-IV	40	2
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Rule of decomposition

STUDENT (Unnormalized table)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
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C2	Oracle&Dev	DEF	100	A1	P-I	20	7
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			105	A6	P-II	30	1
C3	C++	KJP	101	A2	P-II	30	2
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			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

■ One Table

Create a relation with Table identifier of the original table and all the non-repeating attributes.

■ Second Table

The other table contains a copy of the table identifier and all the repeating attributes.

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Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
			101	A2	P-II	30	3
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C2	Oracle&Dev	DEF	100	A1	P-I	20	7
			104	A5	P-III	35	3
			105	A6	P-II	30	1
			101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
			107	A8	P-IV	40	2
			108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

COURSE

Course_Code	Course_Name	Teacher_Name
C1	Visual Basic	ABC
C2	Oracle&Dev	DEF
C3	C++	KJP
C4	Java	Kumar

COURSE_STUDENT

Course_Code	Rollno	Name	System_Used	Hourly_Rate	Total_Hrs
C1	100	A1	P-I	20	7
C1	101	A2	P-II	30	3
C1	102	A3	Celeron	10	6
C1	103	A4	P-IV	40	1
C2	100	A1	P-I	20	7
C2	104	A5	P-III	35	3
C2	105	A6	P-II	30	1
C2	101	A2	P-II	30	2
C3	106	A7	P-IV	40	3
C3	107	A8	P-IV	40	2
C3	108	A9	P-I	20	1
C4	109	A10	Cyrix	20	2

Normalization

Assignment Based on First Normal Form

Assignment Based on First Normal Form

Roll No	Name	Mob No
1	A	M1, M2
2	B	M3, M4
3	C	M5

Assignment Based on First Normal Form

Roll No	Name	Mob No
1	A	M1 M2
2	B	M3 M4
3	C	M5

Solution (Flattening Approach)

Roll No	Name	Mob No
1	A	M1
1	A	M2
2	B	M3
2	B	M4
3	C	M5

Assignment Based on First Normal Form

Roll No	Name	Mob No
1	A	M1 M2
2	B	M3 M4
3	C	M5

Solution (Rule of Decomposition)

Roll No	Name	Mob No
1	A	M1 M2
2	B	M3 M4
3	C	M5

Roll No	Name
1	A
2	B
3	C

Roll No	Mob No
1	M1
1	M2
2	M3
2	M4
3	M5

Normalization

Anomalies of First Normal Form

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

■ Insert Anomaly

Cannot insert the data of Course without Student.

In simple words, cannot insert the data of a course having no student.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

■ Insert Anomaly

Cannot insert the data of Student without Course.

In simple words, cannot insert the data of student who has not registered in any course.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

■ Update Anomaly

Change in Course information requires multiple changes.

For example, to change the name of Course_Code C1 it requires multiple update operations, which is equal to the numBSr of students enrolled in that course.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

■ Update Anomaly

Change in Student information requires multiple changes.

For example, to change the name of RollNo 100 it requires multiple update operations, which is equal to the numBSr of courses enrolled by that student.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

- delete Anomaly

deletion of course information causes deletion of student information too.

This problem is worst for a student who has enrolled in only one course and that course has BSen deleted.

For example, when we delete the data of Course C4 it also causes deletion of record of student with RollNo 109

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

- delete Anomaly

This problem is worst for a student who has enrolled in only one course and that course has BSen deleted.

For example, when we delete the data of Course C4 it also causes deletion of record of student with RollNo 109.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

- delete Anomaly

deletion of student information causes deletion of course information too.

This problem is worst for a course which has only one student enrolled in that course and that student information has BSen deleted.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Anomalies of First Normal Form

- delete Anomaly

This problem is worst for a course which has only one student enrolled in that course and that student information has been deleted.

For example, when we delete the student with RollNo 109 it also causes deletion of record of Course with Course Code C4.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Solution to Solve Anomalies of First Normal Form

- To apply Second Form

Second Normal Form

Second Normal Form

- A relation R is in second normal form (2NF) if and only if it is in 1NF and every non-key attribute is fully functional dependent on the primary key.

Concept of Fully Functional Dependence

Fully Functional Dependence (FFD)

Fully Functional Dependence(FFD) is defined as Attribute Y is FFD on attribute X, if it is FD on X and not FD on any proper subset of X.

$$\frac{(X_1, X_2) \rightarrow Y}{X}$$

$$X_1 \not\rightarrow Y$$

$$X_2 \not\rightarrow Y$$

Fully Functional Dependence (FFD): Example

SNO	PNO	QTY
S1	P1	100
S1	P2	120
S2	P1	100
S2	P2	130

$SNO, PNO \rightarrow QTY$

$SNO \nrightarrow QTY$

$PNO \nrightarrow QTY$

Fully Functional Dependence (FFD)

SNO	SNAME	CITY	STATUS
S1	Sam	Pullman	10
S2	ram	Pullman	10
S3	Sam	del	20
S4	Nik	Seattle	10

$SNO \rightarrow SNAME$ $SNAME \nrightarrow SNO$
 $SNO \rightarrow CITY$ $CITY \nrightarrow SNO$
 $SNO \rightarrow STATUS$ $STATUS \nrightarrow SNO$

Is **$SNO, SNAME \rightarrow CITY$** is FFD?
 $SNO, SNAME \rightarrow CITY$
 $SNO \rightarrow CITY$
 $SNAME \rightarrow CITY$
 $SNO, SNAME \rightarrow CITY$ is not FFD.

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

Process of Conversion to 2NF

STUDENT (Normalized Relation)

Course_ Code	Course_ Name	Teacher_ Name	RollNo	Name	System_ Used	Hourly_ Rate	Total_ Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

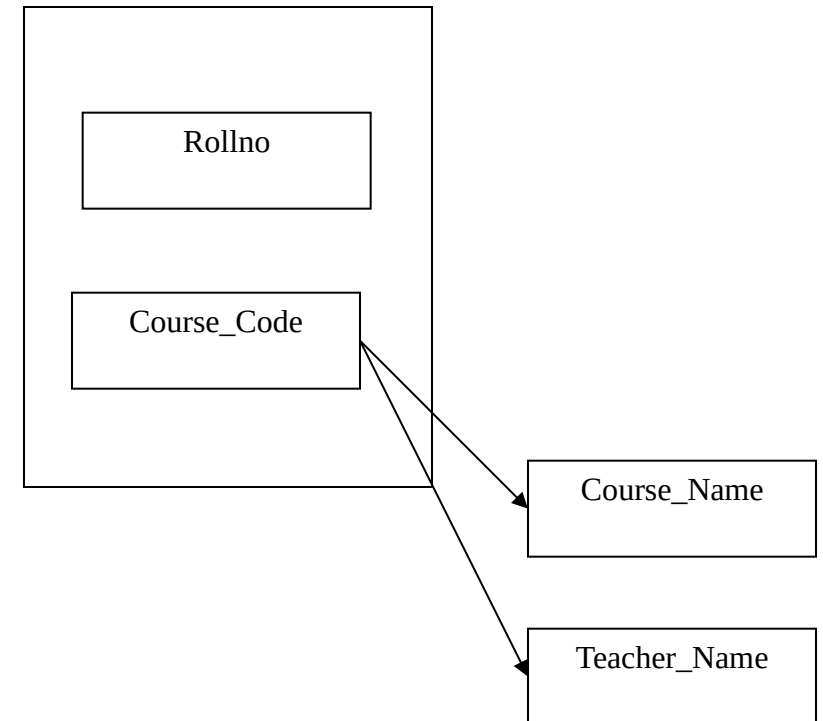
Rollno

Course_Code

Process of Conversion to 2NF

STUDENT (Normalized Relation)

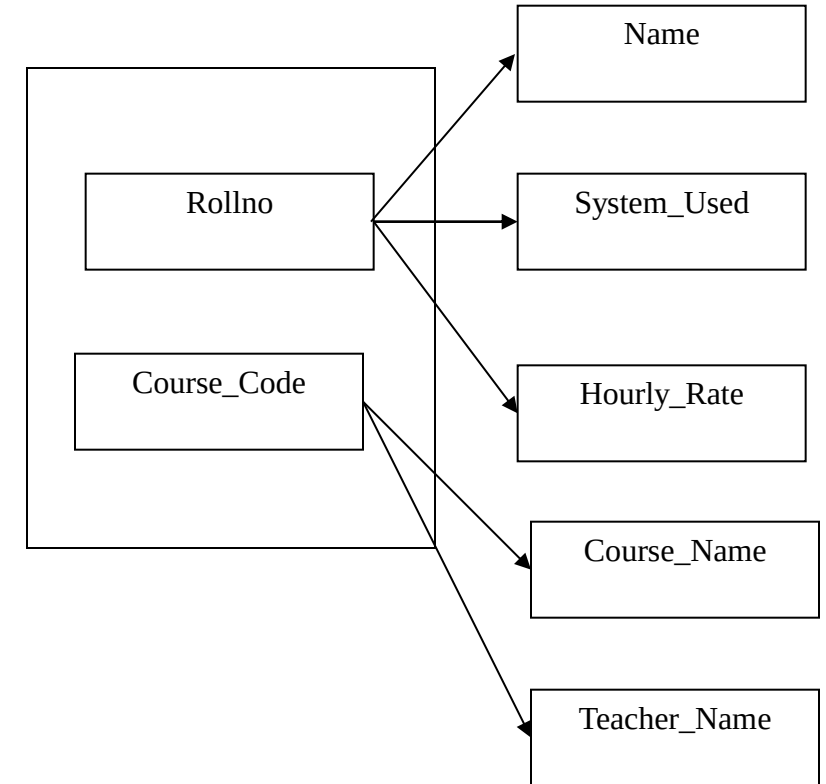
Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2



Process of Conversion to 2NF

STUDENT (Normalized Relation)

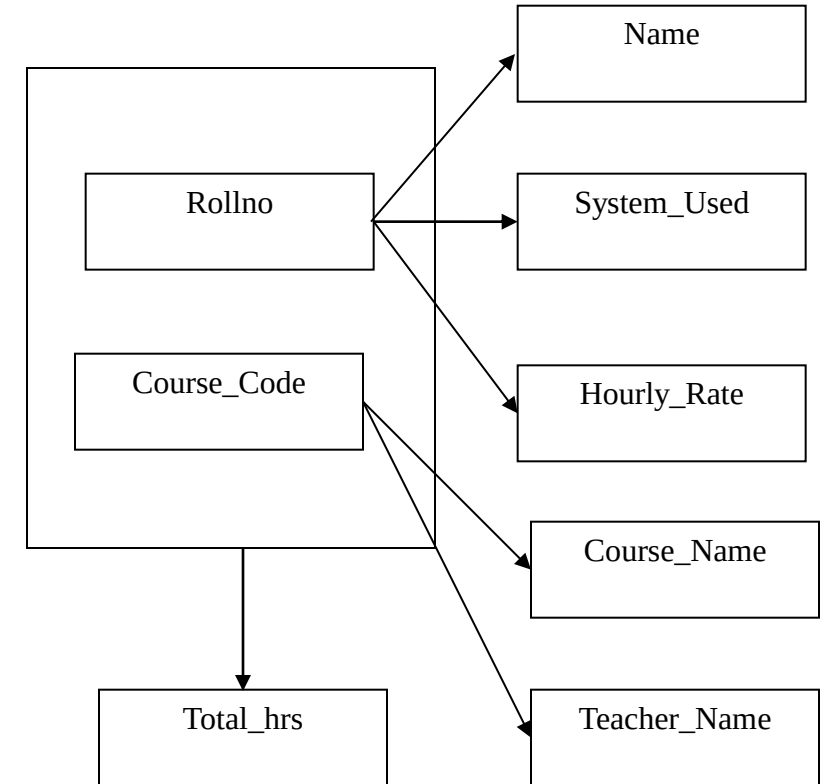
Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2



Process of Conversion to 2NF

STUDENT (Normalized Relation)

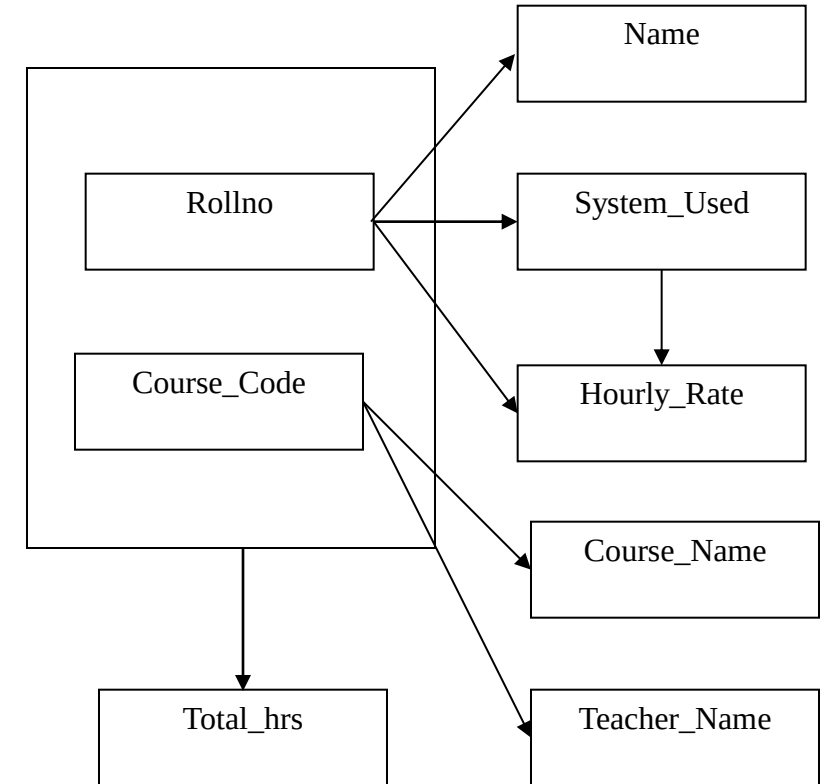
Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2



Process of Conversion to 2NF

STUDENT (Normalized Relation)

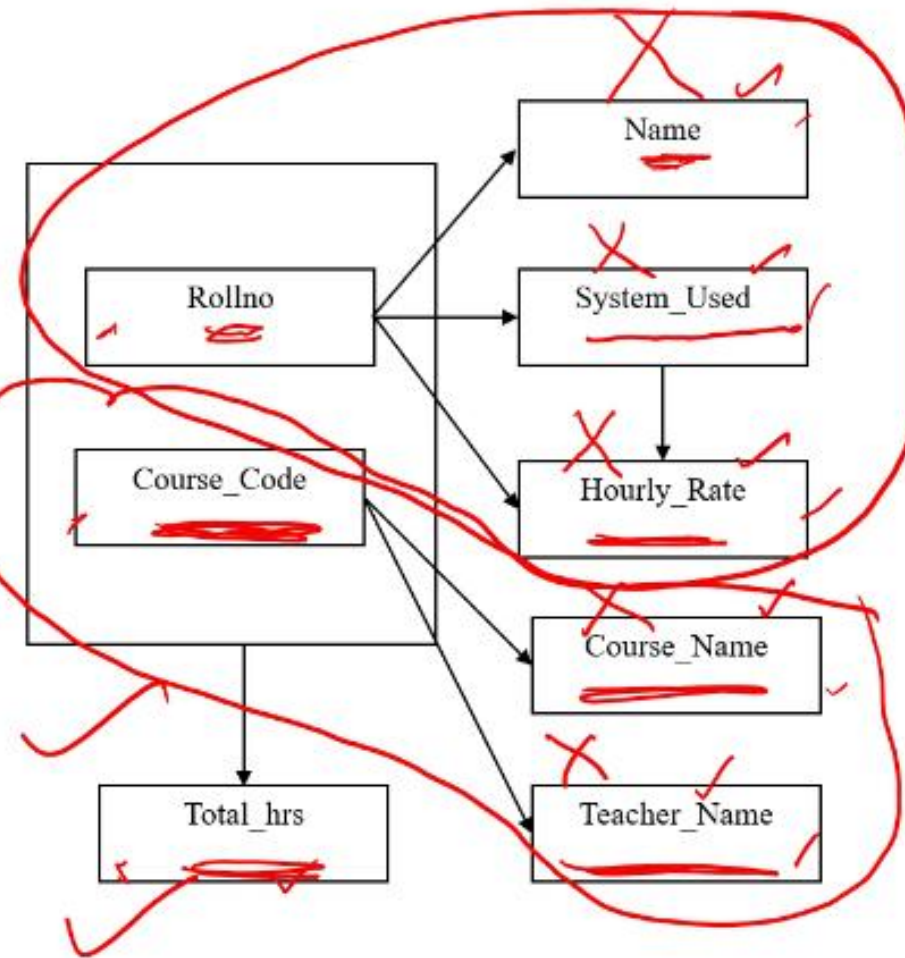
Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2



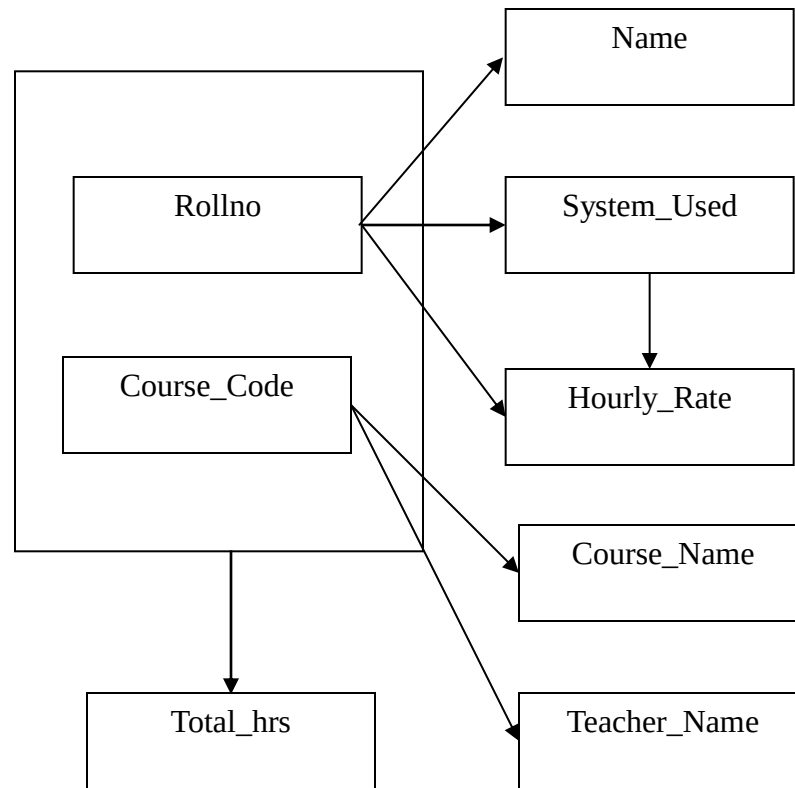
Process of Conversion to 2NF

STUDENT (Normalized Relation)

Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

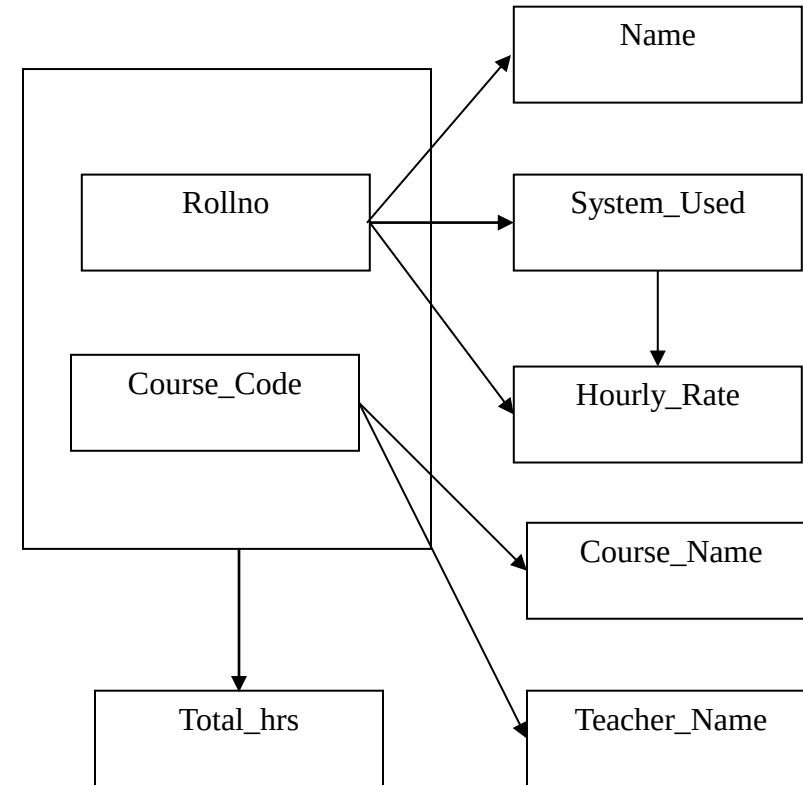


FD Diagram



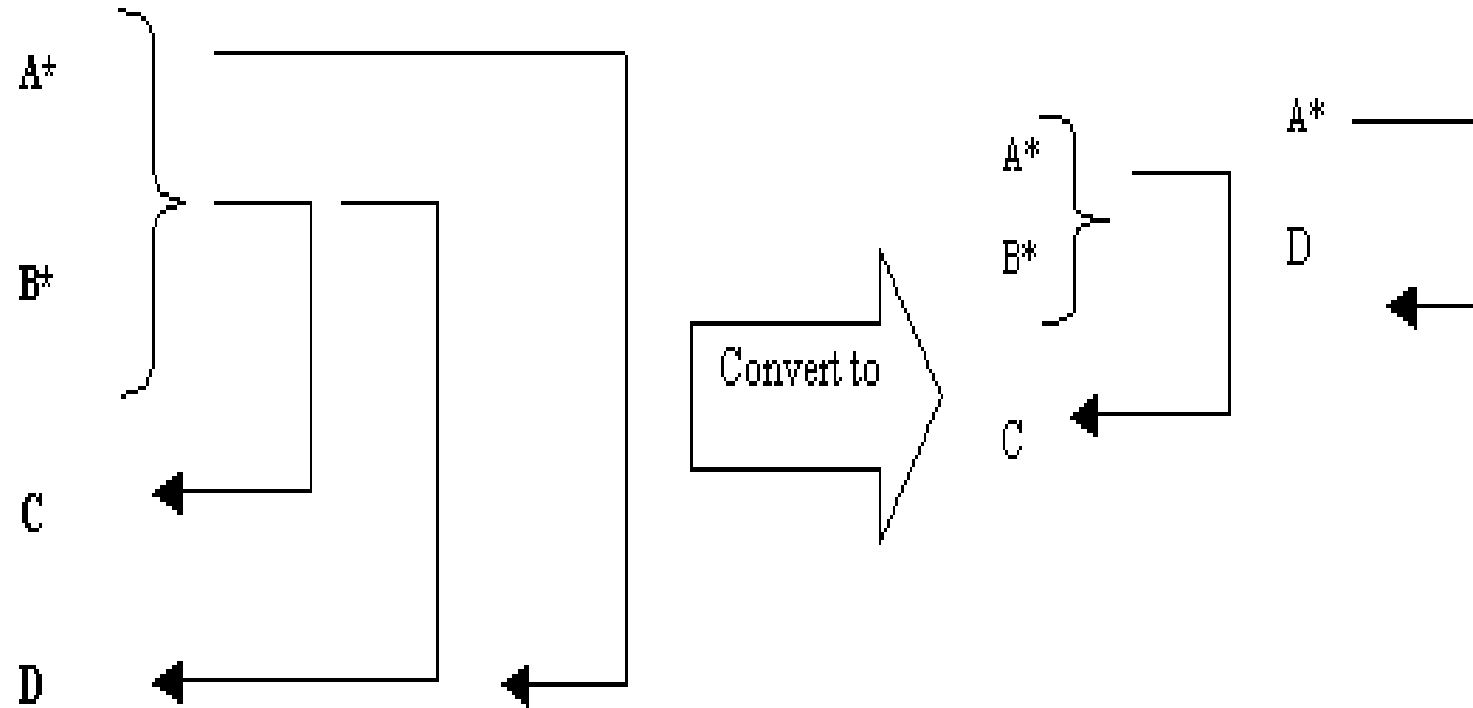
Applying 2NF Rule

- A relation R is in second normal form (2NF) if and only if it is in 1NF and every non-key attribute is fully functional dependent on the primary key.

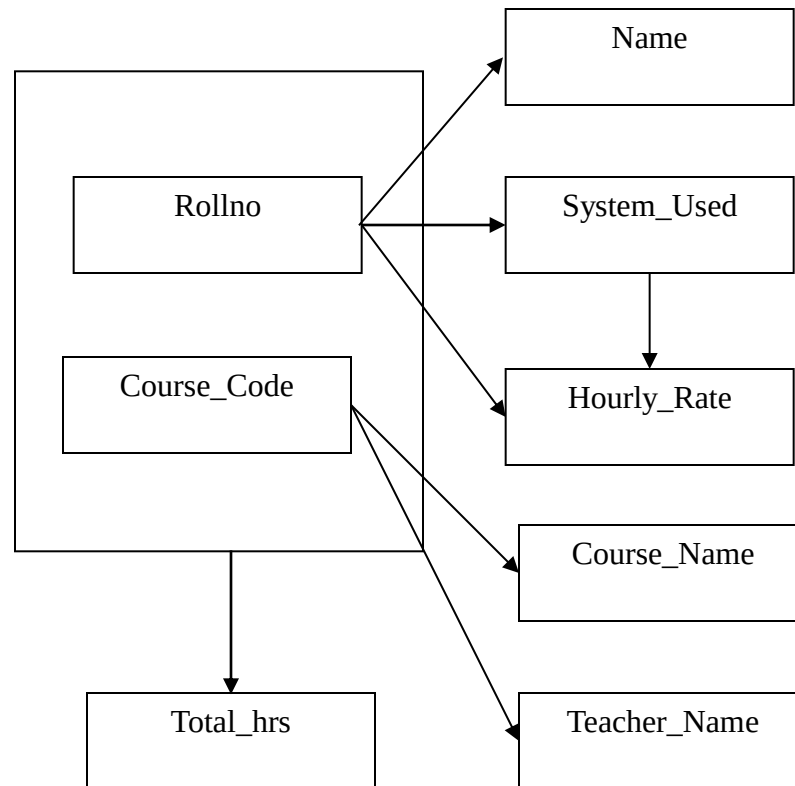


Second Normal Form Rule of Decomposition

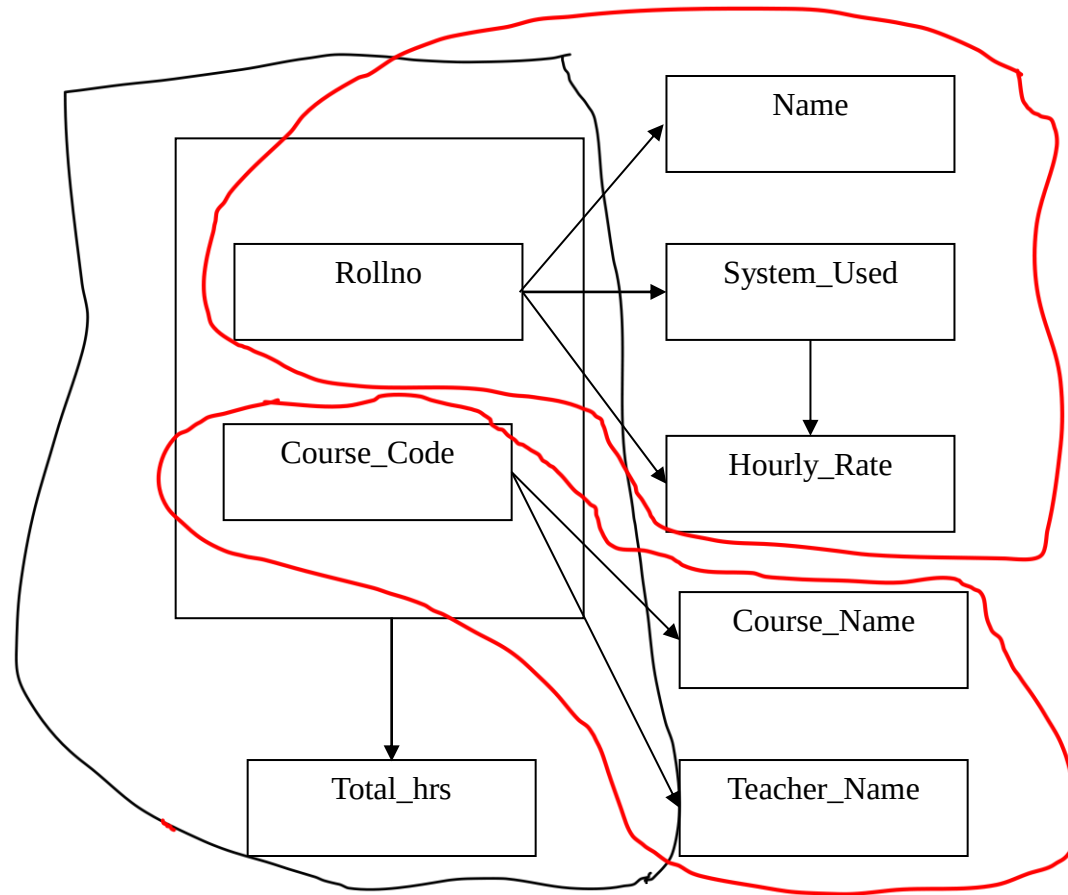
Rule of Decomposition for 2NF



FD Diagram



Rule of Decomposition for 2NF



Source Table

STUDENT (Normalized Relation)

Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

STUDENT_SYSTEM_CHARGE

Rollno	Name	System_Used	Hourly_Rate
100	A1	PentiumI	20
101	A2	PentiumII	30
102	A3	Celeron	10
103	A4	PentiumIV	40
104	A5	P-III	35
105	A6	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

COURSE

Course_Code	Course_Name	Teacher_Name
C1	Visual Basic	ABC
C2	Oracle&Dev	DEF
C3	C++	KJP
C4	Java	Kumar

HOURS_ASSIGNED

Course_Code	RollNo	Total_Hrs
C1	100	7
C1	101	3
C1	102	6
C1	103	1
C2	100	7
C2	104	3
C2	105	1
C2	101	2
C3	106	3
C3	107	2
C3	108	1
C4	109	2

Is it Normalized

Course	Content
Programming	Java, c++
Web	HTML, PHP, ASP

First Normal Form

Course	Content
Programming	Java
Programming	C++
Web	HTML
Web	PHP
Web	ASP



Anomalies of Second Normal Form

STUDENT (Normalized Relation)

Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

COURSE

Course_Code	Course_Name	Teacher_Name
C1	Visual Basic	ABC
C2	Oracle&Dev	DEF
C3	C++	KJP
C4	Java	Kumar

STUDENT_SYSTEM_CHARGE

Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

HOURS_ASSIGNED

Course_Code	RollNo	Total_Hrs
C1	100	7
C1	101	3
C1	102	6
C1	103	1
C2	100	7
C2	104	3
C2	105	1
C2	101	2
C3	106	3
C3	107	2
C3	108	1
C4	109	2

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20
		Laptop	40

- Insert Anomaly
- Since Rollno is PK, so cannot insert the data of System_Used without Student
- In Simple words, cannot insert the Hourly_rate of Laptop as there is no student is using System Laptop.

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

- Update Anomaly
- Updation of Hourly_Rate of System requires multiple update operations.

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

- Update Anomaly
- For example, if we have to change the Hourly_Rate of System P-I from 20 to 30 we have to make multiple changes, which is equal to the numBSr of students using P-I

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

- delete Anomaly
- The deletion of Student data causes the deletion of System data too.
- This problem is worst for a system which is used by only one student.

Anomalies in STUDENT_SYSTEM_CHARGE Table

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

- delete Anomaly
- This problem is worst for a system which is used by only one student and who data is removed.
- For example, if we remove the data of student with RollNo 102 then we also lose the Hourly_Rate information of System_Used Celeron

Solution to Solve these anomalies

- To apply third normal form.

STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

Third Normal Form

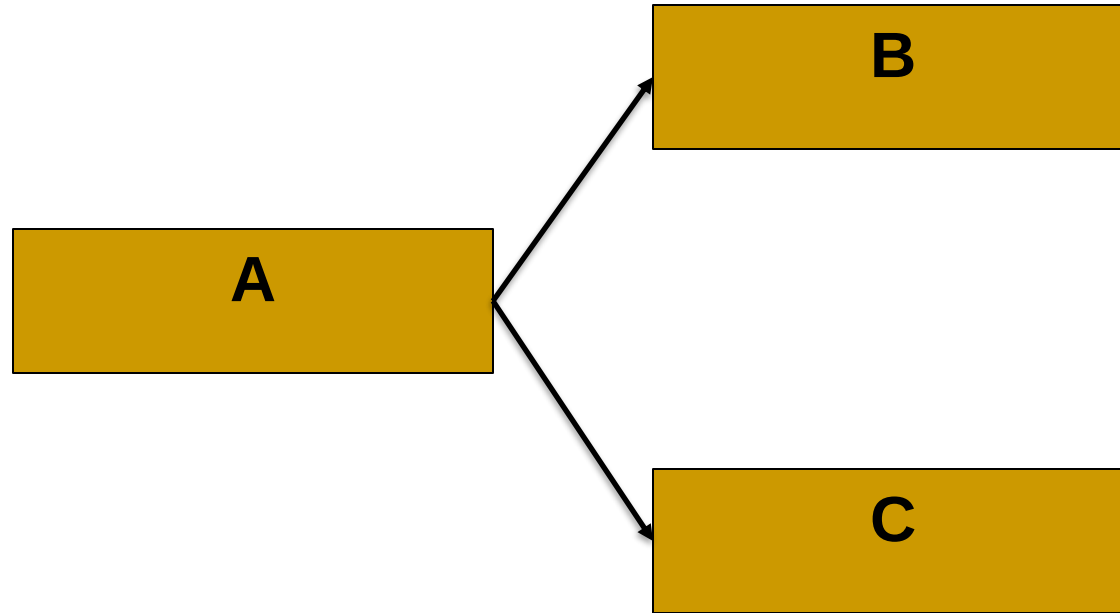
Third Normal Form

A relation R is in Third Normal Form (3NF) if and only if the following conditions are satisfied simultaneously:

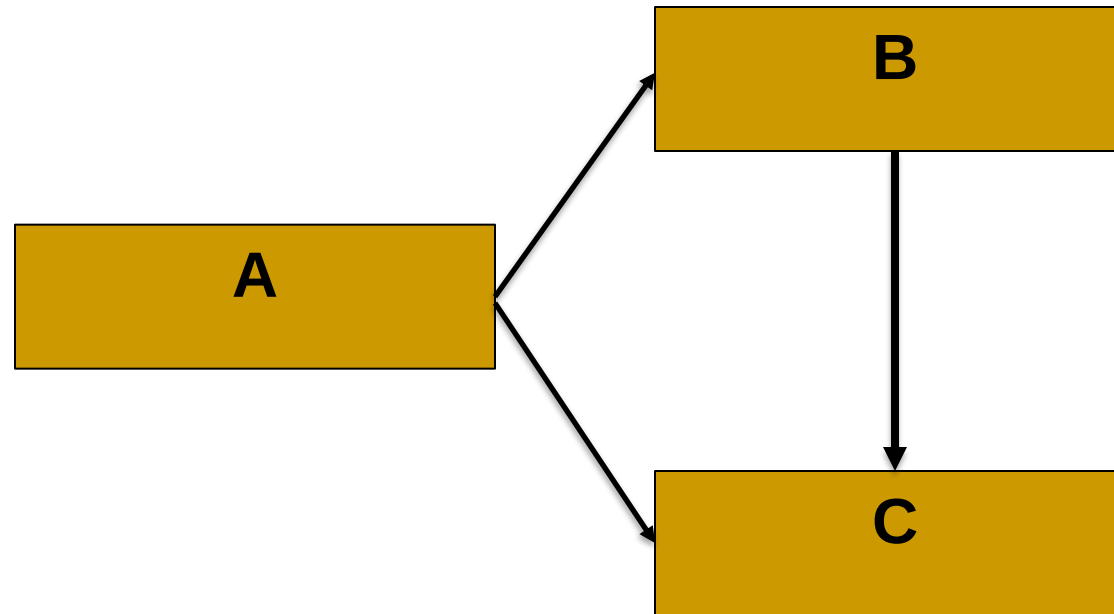
- R is already in 2NF
- No nonprime attribute is transitively dependent on the key.

Another way of expressing the conditions for Third Normal Form is as follows:

- R is already in 2NF
 - No nonprime attribute functionally determines any other nonprime attribute.
-

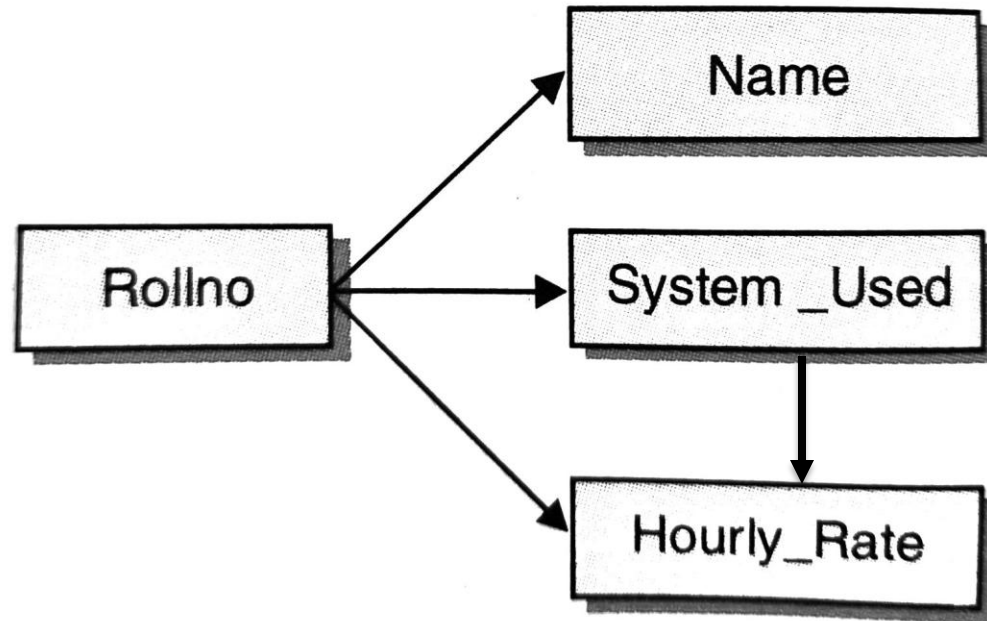


Concept of Transitive Dependence

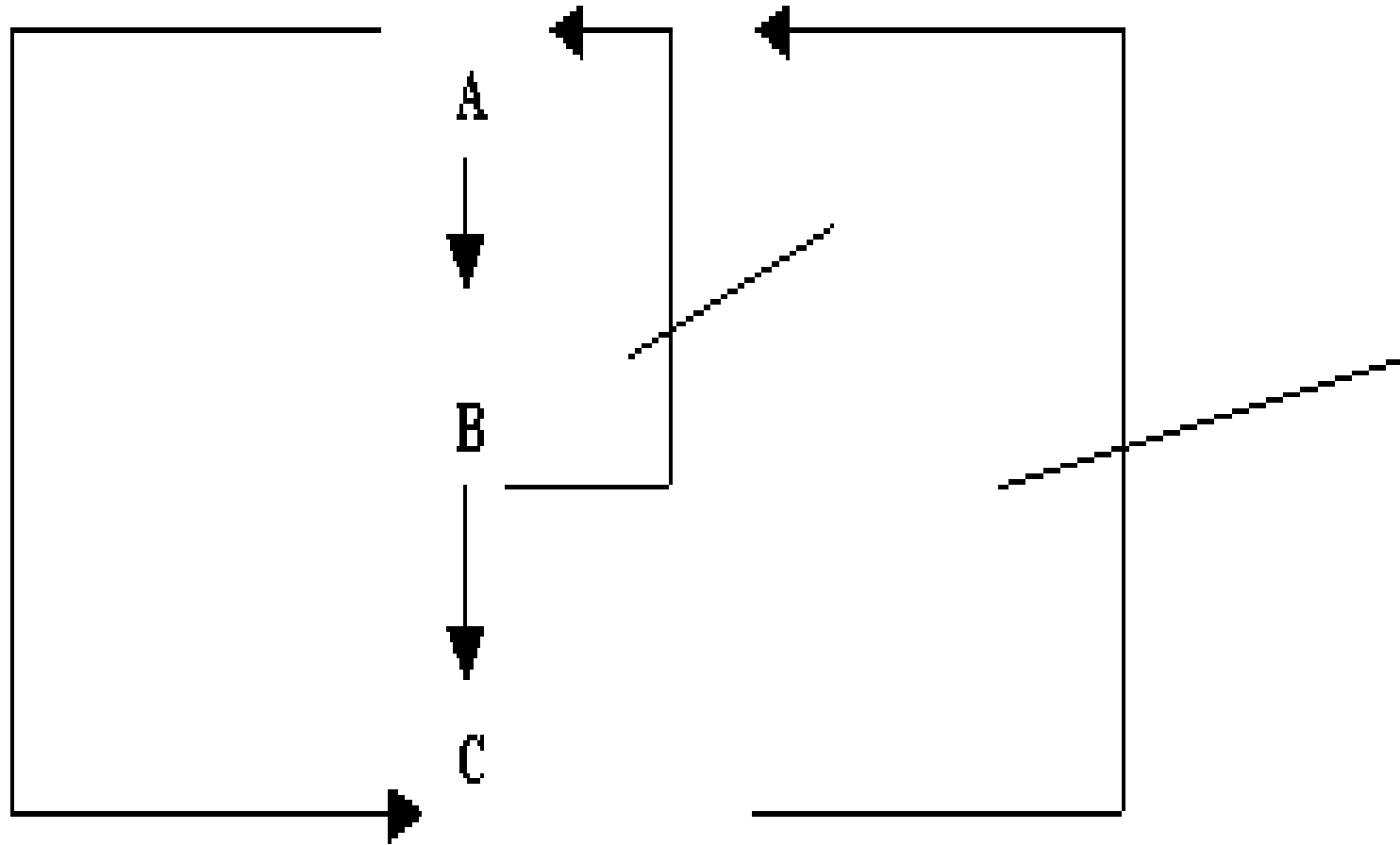


- C is dependent on PK A two ways: one directly and other indirectly through B.
- This indirect path of B between A and C through B causes Transitive Dependence of C on A.

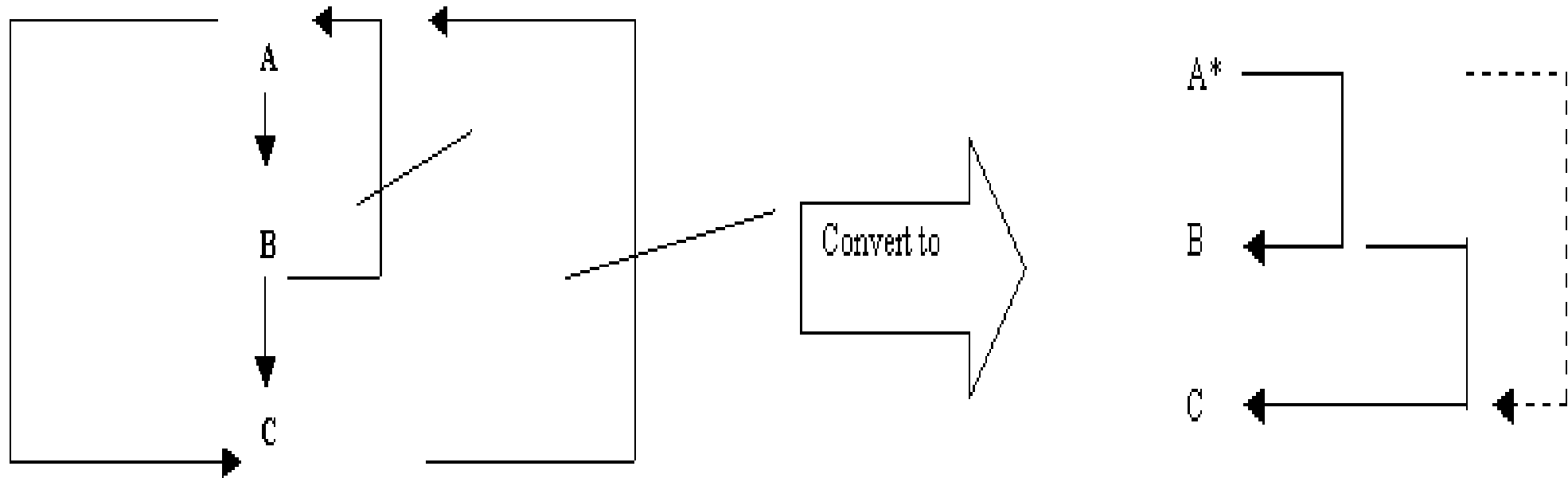
Concept of Transitive Dependence



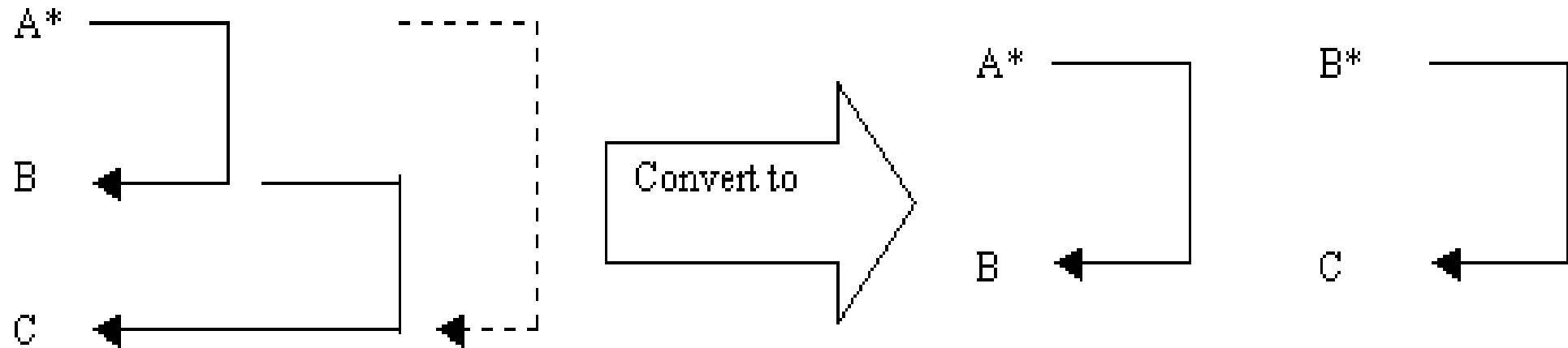
Concept of Transitive Dependence



Concept of Transitive Dependence



Role to Convert a Relation to Third Normal Form



STUDENT_SYSTEM_CHARGE			
Rollno	Name	System_Used	Hourly_Rate
100	A1	P-I	20
101	A2	P-II	30
102	A3	Celeron	10
103	A4	P-IV	40
104	A5	P-II	30
106	A7	P-IV	40
107	A8	P-IV	40
108	A9	P-I	20
109	A10	Cyrix	20

STUDENT_STSTEM		
Rollno	Name	System_Used
100	A1	P-I
101	A2	P-II
102	A3	Celeron
103	A4	P-IV
100	A1	P-I
104	A5	P-III
105	A6	P-II
101	A2	P-II
106	A7	P-IV
107	A8	P-IV
108	A9	P-I
109	A10	Cyrix

CHARGES	
System_Used	Hourly_Rate
P-I	20
P-II	30
Celeron	10
P-IV	40
P-III	35
Cyrix	20

STUDENT (Normalized Relation)

Course_Code	Course_Name	Teacher_Name	RollNo	Name	System_Used	Hourly_Rate	Total_Hrs.
C1	Visual Basic	ABC	100	A1	P-I	20	7
C1	Visual Basic	ABC	101	A2	P-II	30	3
C1	Visual Basic	ABC	102	A3	Celeron	10	6
C1	Visual Basic	ABC	103	A4	P-IV	40	1
C2	Oracle&Dev	DEF	100	A1	P-I	20	7
C2	Oracle&Dev	DEF	104	A5	P-III	35	3
C2	Oracle&Dev	DEF	105	A6	P-II	30	1
C2	Oracle&Dev	DEF	101	A2	P-II	30	2
C3	C++	KJP	106	A7	P-IV	40	3
C3	C++	KJP	107	A8	P-IV	40	2
C3	C++	KJP	108	A9	P-I	20	1
C4	Java	Kumar	109	A10	Cyrix	20	2

COURSE

Course_Code	Course_Name	Teacher_Name
C1	Visual Basic	ABC
C2	Oracle&Dev	DEF
C3	C++	KJP
C4	Java	Kumar

STUDENT_STSTEM

Rollno	Name	System_Used
100	A1	P-I
101	A2	P-II
102	A3	Celeron
103	A4	P-IV
100	A1	P-I
104	A5	P-III
105	A6	P-II
101	A2	P-II
106	A7	P-IV
107	A8	P-IV
108	A9	P-I
109	A10	Cyrix

CHARGES

System_Used	Hourly_Rate
P-I	20
P-II	30
Celeron	10
P-IV	40
P-III	35
Cyrix	20

HOURS_ASSIGNED

Course_Code	RollNo	Total_Hrs
C1	100	7
C1	101	3
C1	102	6
C1	103	1
C2	100	7
C2	104	3
C2	105	1
C2	101	2
C3	106	3
C3	107	2
C3	108	1
C4	109	2

Boyce/Codd N/F (BCNF)

BCNF

- BCNF is simply a stronger definition of 3NF. BCNF makes no explicit reference to first and second normal form as such, nor the concept of full and transitive dependence.
 - BCNF states that
 - A relation R is in Boyce/Codd N/F (BCNF) if and only if every determinant is a candidate key.
-

BCNF

- Determinant is a simple attribute or composite attribute on which some other attribute is fully functionally dependent.
- For example Qty is FFD on (Sno, Pno)
- $(\text{Sno}, \text{Pno}) \rightarrow \text{Qty}$, here
- (Sno, Pno) is a composite determinant.
- $\text{Sno} \rightarrow \text{Sname}$
- Here Sno is simple attribute determinat.

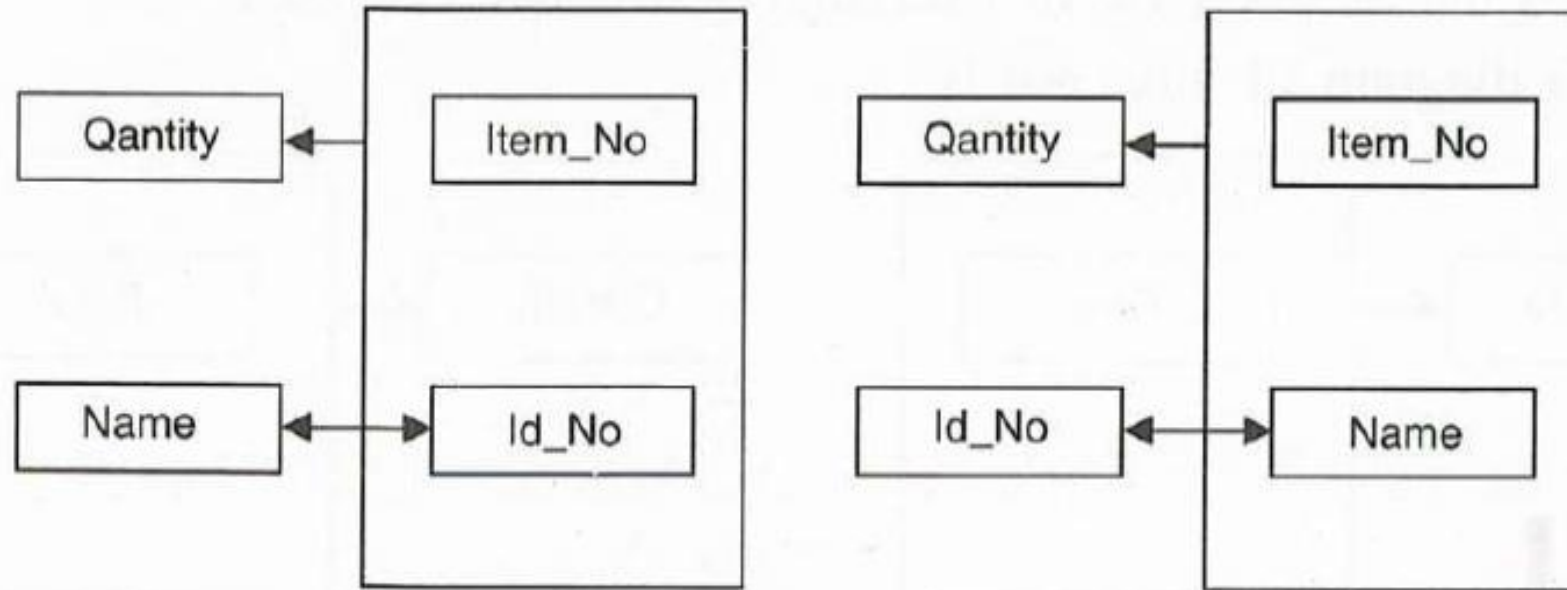
BCNF

- **Differences in 3NF and BCNF**
- In order to show the difference between 3NF and BCNF, relations having overlapping of candidate keys are considered in detail.



Manufacturer

Id_No	Name	Item_No	Quantity
M101	Electronics USA	H3772	1000
M101	Electronics USA	J08732	700
M101	Electronics USA	Y23490	200
M322	Electronics-R-Us	H3772	900



BCNF

- **Overlapping of Candidate keys:**
- Two candidate keys overlap if they involve two or more attributes each and have an attribute in common.
- $(Id_no, Item_No) \rightarrow Quantity$
- $(Name, Item_No) \rightarrow Quantity$
- $Item_No \rightarrow Name$
- $Name \rightarrow Item_No$

Fourth Normal Form

Fourth Normal Form (4NF)

A relation R is in Fourth Normal Form (4NF) if and only if the following conditions are satisfied simultaneously:

- R is already in 3NF or BCNF.
- If it contains no multi-valued dependencies.

Multi-Valued Dependency (MVD)

- MVD can be defined informally as follows:
- It means that if in a relation R having A, B and C as attributes, B and C are multi-value facts about A, which is represented as $A \twoheadrightarrow B$ and $A \twoheadrightarrow C$, then multi value dependency exist only if B and C are independent of each other.
- $A \twoheadrightarrow B$
- $A \twoheadrightarrow C$
- B and C should be independent to each other.

Course_Code	Student_Name	Book_Name
DBMS	Sam	Simplified Approach to DBMS
	ram	Concepts of DBMS
		DBMS With Oracle
Machine Learning	ram	Simplified Approach to ML
	SHAM	Concepts of ML
	SamA	
Advanced DBMS	Sam	Simplified Approach to DBMS
		Databases

Course_Code	Student_Name	Book_Name
DBMS	Sam	Simplified Approach to DBMS
DBMS	Sam	Concepts of DBMS
DBMS	Sam	DBMS With Oracle
DBMS	ram	Simplified Approach to DBMS
DBMS	ram	Concepts of DBMS
DBMS	ram	DBMS With Oracle
Machine Learning	ram	Simplified Approach to ML
Machine Learning	ram	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	SamA	Simplified Approach to ML
Machine Learning	SamA	Concepts of ML
Advanced DBMS	Sam	Simplified Approach to DBMS
Advanced DBMS	Sam	Databases

Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases

- Table is 1NF.
- Primary Key is
- Combination of
 - Course_Code
 - Student_Name
 - Book_Name

Insert Anomaly

Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases

- Cannot insert the data of course and book without student
- Cannot insert the data of course and student without book
- In order to add a new book for a course, we have to make multiple entries which is equal to numBSr of students taking that course.
- In order to add a new student for a course, we have to make multiple entries which is equal to numBSr of books for that course.

Update Anomaly

- Student name updation needs multiple changes and similarly book name updation needs multiple changes

Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases

delete Anomaly

Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases

- When we delete the data of student it BScouse deletion of course and book information
- When we delete the data of course it causes deletion of student and book information
- When we delete the data of book it cause deletion of course and student information

Lets Normalize this database

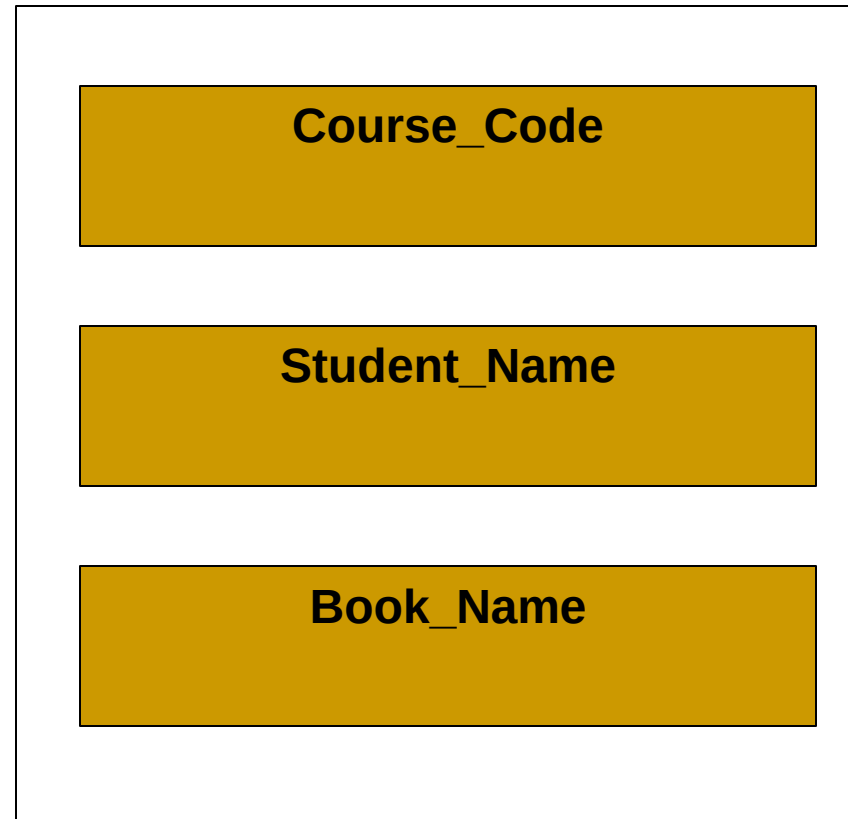
Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases

Course_Code

Student_Name

Book_Name

Lets Normalize this database



Lets Normalize this database

Course_Code

Student_Name

Book_Name

- Lets Apply different normal forms tests over this FD
- 1 NF
- 2 NF
- 3 NF
- BCNF

Course_Code

Student_Name

Book_Name

Lets Apply 4NF

Course_Code

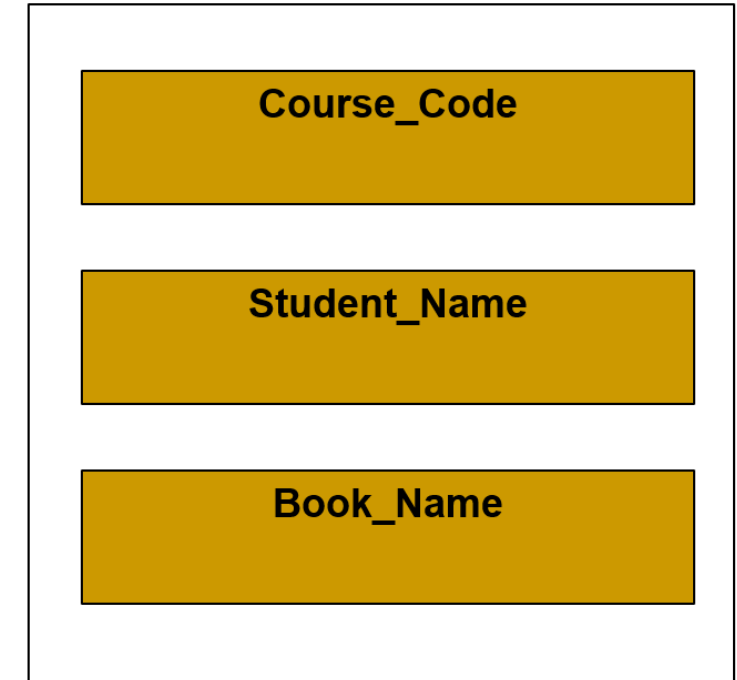
Student_Name

Book_Name

- $A \twoheadrightarrow B$
- $A \twoheadrightarrow C$
- B and C should be independent to each other.
- $\text{Course_Code} \twoheadrightarrow \text{Student_name}$
- $\text{Course_Code} \twoheadrightarrow \text{Book_Name}$
- Student_name and Book_Name are independent to each other.

Lets Apply 4NF

Course_Code	Student_Name	Book_Name
DBMS	RAJ	Simplified Approach to DBMS
DBMS	RAJ	Concepts of DBMS
DBMS	RAJ	DBMS With Oracle
DBMS	RAM	Simplified Approach to DBMS
DBMS	RAM	Concepts of DBMS
DBMS	RAM	DBMS With Oracle
Machine Learning	RAM	Simplified Approach to ML
Machine Learning	RAM	Concepts of ML
Machine Learning	SHAM	Simplified Approach to ML
Machine Learning	SHAM	Concepts of ML
Machine Learning	RAJA	Simplified Approach to ML
Machine Learning	RAJA	Concepts of ML
Advanced DBMS	RAJ	Simplified Approach to DBMS
Advanced DBMS	RAJ	Databases



- Course_Code →→ Student_name
- Course_Code →→ Book_Name
- Student_name and Book_Name are independent to each other.

Rule to transform a relation into Fourth Normal Form

- A relation R having A,B, and C, as attributes can BS non loss-decomposed into two projections R1(A,B) and R2(A,C) if and only if the MVD $A \twoheadrightarrow B|C$ hold in R.
- $\text{Course_Code} \twoheadrightarrow \text{Student_name}|\text{Book_Name}$

Decompose this to 4NF

- $\text{Course_Code} \twoheadrightarrow \text{Student_name}$
- $\text{Course_Code} \twoheadrightarrow \text{Book_Name}$
- To put it into 4NF, two separate tables are formed as shown below:
- $\text{COURSE_STUDENT}(\text{Course_Code}, \text{Student_name})$
- $\text{COURSE_BOOK}(\text{Course_Code}, \text{Book_Name})$

Final Tables in 4NF

Course_Code	Student_Name
DBMS	Sam
DBMS	ram
Machine Learning	ram
Machine Learning	SHAM
Machine Learning	SamA
Advanced DBMS	Sam

Course_Code	Book_Name
DBMS	Simplified Approach to DBMS
DBMS	Concepts of DBMS
DBMS	DBMS With Oracle
Machine Learning	Simplified Approach to ML
Machine Learning	Concepts of ML
Advanced DBMS	Simplified Approach to DBMS
Advanced DBMS	Databases

5NF

To **complete the discussion and further scope**, you can always **decompose** a relation **R** if it is already in 4NF and it still suffers from Insert, Update and Delete anomalies into **R1**, **R2**, and **R3** provided that after **joining them**, you get the **original table R back**. This property is known as **Projection-Join Dependency (PJD)**.

5NF

Key Points About Projection-Join Dependency (PJD):

- It ensures that **lossless decomposition** is maintained.
- Even after breaking a relation into smaller relations, no data is lost, and the original table can be **reconstructed using joins**.
- It is a fundamental concept in **database normalization**, ensuring **redundancy reduction** while maintaining **data integrity**.

For more information

- Please visit: www.parteeekbhatia.com
 - For Feedback: parteeek.bhatia@gmail.com
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 - For Workshops and invited talks, please write to:
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-

Thanks
